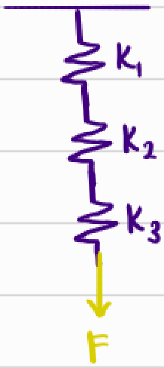
Springs

Hooke Law

- Deformation of spring, x over elastic range $\propto$ Applied force via spring constant, k
- $F = kx$

Series [F on spring is same]

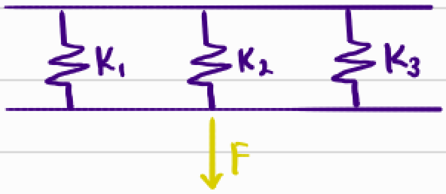


$$\boxed{\Sigma x = x_1 + x_2 + x_3}$$

$$\Longrightarrow \Sigma \frac{F}{k} = \frac{F_1}{k_1} + \frac{F_2}{k_2} + \frac{F_3}{k_3}$$

$$\Longrightarrow \boxed{\Sigma \frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3}}$$

Parallel [x of spring is same]

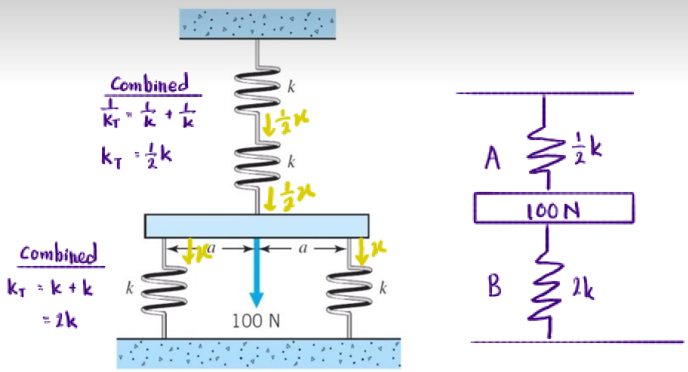


$$\boxed{\Sigma F = F_1 + F_2 + F_3}$$

$$\Longrightarrow \Sigma kx = (kx)_1 + (kx)_2 + (kx)_3$$

$$\boxed{\Sigma k = k_1 + k_2 + k_3}$$

A bar of weight 100 N is support by 4 springs in the arrangement shown in the figure below. The springs have the same stiffness of k. Determine the force carried in each spring.



$\therefore$ $100\text{N} = F_A + F_B$

$100 = \frac{1}{2}kx + 2kx$ [Magnitude of x must be same, since the bar can't be at 2 places at once. Extension of top springs = Contraction of bottom springs. Let deformation be x]

$2.5kx = 100$

$x = \frac{40}{k}$

$\therefore$ Bottom spring $= kx$
$= k\left(\frac{40}{k}\right)$
$= 40\text{N}$

Top spring $= k\left(\frac{x}{2}\right)$
$= k\left(\frac{20}{k}\right)$
$= 20\text{N}$

Rmb $\Sigma x = x_1 + x_2$
Since bottom only moved down x, top only can move down total of x also

Pulley

Mechanical Advantage, MA

- Ratio of F_{output} to the F_{input}, $\frac{F_{output}}{F_{input}}$
- A measure of how much a machine amplifies input force at the cost of increasing the distance over which the force is applied [Work is conserved, Energy is conserved]
- Since work [$W = Fd$] is conserved, $F \propto \frac{1}{d}$

$$\frac{W/d_{out}}{W/d_{in}} = \frac{d_{in}}{d_{out}}$$

Example: I want to lift 100N object by 1 meter

With MA = 1 Pulley

- I need to apply 100N of force over 1 meter

$W = 100(1) = 100\,J$

With MA = 4 Pulley

- I only need to apply 25 N of force but pull 4 meters of rope

$W = 25(4) = 100\,J$

Compound Pulley [Consists of fixed & movable pulleys]

→ means : Applies force on

Mechanism : • "Splits" the rope w̄ pulleys so that a rope acts like a rope in parallel, thus sharing the load like springs in parallel.

• Load → Each pulley → Rope (Parallel) ← Force — To create tension

• Thus, Force applied causes tension on every segment of the rope

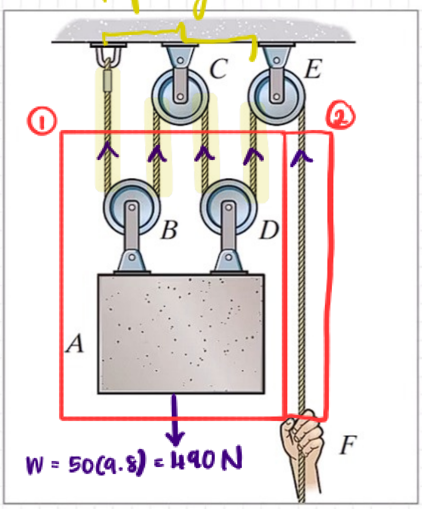
↳ Effective total tension = weight of load.

• This allows us to use less force bcuz our force will be multiplied by the no. of rope segment (MA). And this multiplied force will be used to lift the object

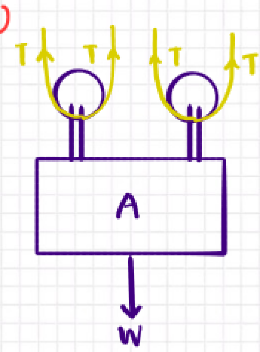
• Since the rope has to go around the pulley rather a straight rope, more rope have to be pulled for the rope to actually pull on the load.

"Parallel" Pulleys [Segment of strings are parallel]

Determine the force F required to keep block A stationary if it weighs 50 kg. Also, calculate the mechanical advantage in this system and the distance the cable needs to be pulled down to raise block A by 1 m. Assume the gravitation acceleration is $g = 9.8\ \text{m/s}^2$.



①

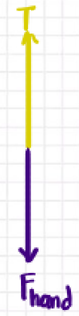


$\Sigma F_y = 0$

$4T - mg = 0$

$T = 122.5\ \text{N}$

②



$\Sigma F_y = 0$

$T - F = 0$

$F = 122.5\ \text{N}$

$\therefore MA = \dfrac{490}{122.5}$

$= 4$

$\therefore MA = \dfrac{d_{in}}{d_{out}}$

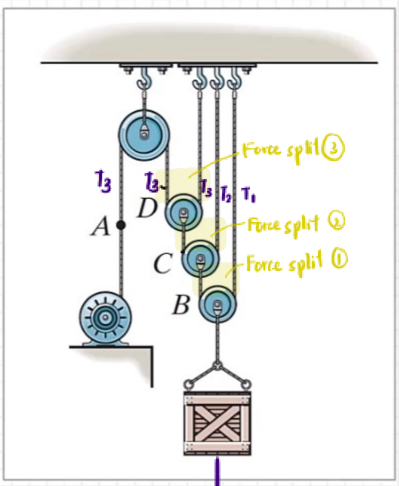
$\boxed{MA = \dfrac{F_{out}}{F_{in}} = \dfrac{W/d_{out}}{W/d_{in}} = \dfrac{d_{in}}{d_{out}}}$

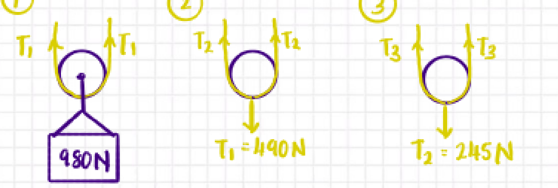
$4 = \dfrac{d_{in}}{1}$

$d_{in} = 4\ \text{m}$

"In Series" Pulley [Segment of strings splits to top and bottom]

The pulley system below needs to support a 100 kg load in equilibrium. Four motors are available with a pulling force of 100 N, 200 N, 300 N, and 400 N. The cost of the motors increase with increased pulling force capacity. Determine the cheapest motor that would service this pulley system. Assume the gravitation acceleration is $g = 9.8\ \text{m/s}^2$.





$2T_1 = 980\text{N}$

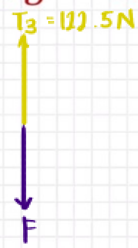
$T_1 = 490\text{N}$

$2T_2 = 490\text{N}$

$T_2 = 245\text{N}$

$2T_3 = 245$

$T_3 = 122.5\text{N}$



$F = 122.5\text{N}$

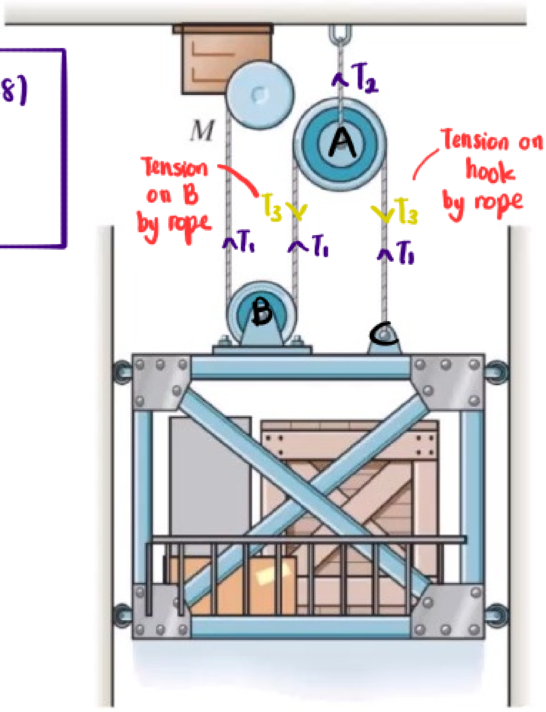
Consider the pulley system below. A freight elevator is lifted using a pulley system and a motor. If the elevator and cargo weigh a combined total of 1000 kg then determine the required tension in the cable connected to the motor to keep it in equilibrium. Also, determine the tension in the cable connected to the pulley fixed to the ceiling. Assume that the wheels are on frictionless tracks. Take the gravitational acceleration to be $g = 9.8\ \mathrm{m/s^2}$.

① "Parallel"

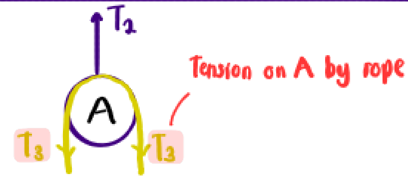
$$\Sigma F_y = 3T_1 - 1000(9.8)$$

$$3T_1 = 98{,}000$$

$$T_1 = 3266.667\ \text{N}$$



② "In series"



$|T_3| = |T_1| \Longrightarrow$ Newton 3rd Law

$$\Sigma F_y = T_2 - 2T_3$$

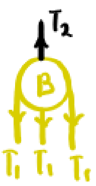
$$0 = T_2 - 2T_1$$

$$0 = T_2 - 2(3266.667)$$

$$T_2 = 6533.3333\text{N}$$

Consider the elevator system shown in the figure below. Determine the force P required to lift the 1000 kg elevator. Also, determine the mechanical advantage of this pulley system. Consider the gravitational acceleration $g = 9.8\ \mathrm{m/s^2}$.

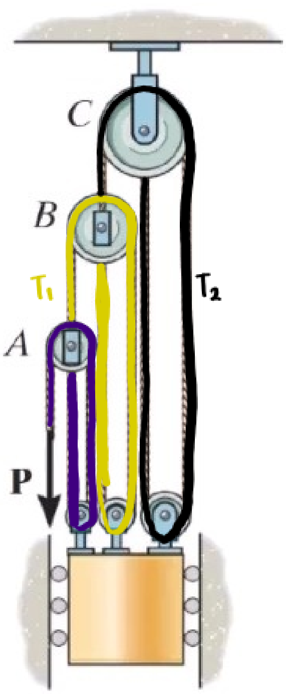
$T_1 = 3P$ — ①

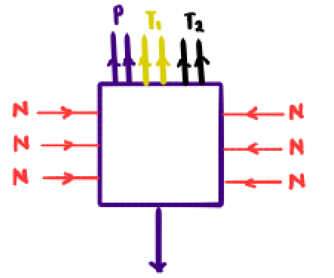


$T_2 = 3T_1$ — ②

① → ②

$T_2 = 9P$





$$W = 1000(9.8) = 9800$$

$$\Sigma F_y = 2P + 2T_1 + 2T_2$$

$$9800 = 2P + 2(3P) + 2(9P)$$

$$P = 376.9231\ \text{N} \#$$

$$\therefore \text{MA} = \frac{9800}{376.9231} = 26 \#$$